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Does Adult Digital Learning Pay Off? The Impact on Labor Market Returns

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ABSTRACT

This paper investigates the labor market returns to adult digital learning. We find that digital learning generates a 6.9% increase in labor income, with sustained positive dynamic effects. Building on a theoretical model that incorporates online training, we empirically demonstrate that digital learning both accumulates human capital and reduces learning costs through flexible, fragmented learning. Unlike traditional offline education, digital learning can overcome implicit gender and age barriers, particularly boosting earnings in nonphysical occupations. Our findings offer important policy insights for on-the-job training and human capital development in developing countries and aging societies.

JEL Classification: I25, I26, J24

1 | Introduction

The exponential development of digital technologies is reshaping global labor markets (Acemoglu and Restrepo 2018). Numerous literatures demonstrate the negative effect of digital technologies on both manual and skilled labor (Acemoglu and Restrepo 2019; Felten et al. 2023). However, it ignores the impact of digital technologies on learning, especially for vocational training.

Vocational training aims to equip workers with the necessary skills to remain competitive in a rapidly changing job market (Murphy and Topel 2016). However, traditional forms of vocational training, which typically rely on self-study or offline, classroom-based education, face several challenges. These include the high opportunity cost of learning, long periods of time away from the workplace, and a growing mismatch between workers' needs and the quality of training programs (Armona et al. 2018).

Therefore, the usefulness of the role of vocational training is still being debated. Some studies believe that vocational training can significantly increase wages (Jacobson et al. 2005; Böckerman et al. 2019; Bratsberg et al. 2020), while others express concerns about the effectiveness (Schwerdt et al. 2012; Armona et al. 2018; Hogendoorn et al. 2019). However, previous studies mainly study the effect of traditional offline vocational training and ignore the effect of digital technology.

As digital technologies evolve, digital learning has emerged as a new and promising avenue for on-the-job training. Digital learning leverages online platforms to enable workers to acquire knowledge at their convenience, breaking down geographical and time barriers. These platforms offer significant advantages over traditional offline methods. Digital learning reduces the cost of learning by providing access to free or low-cost resources that were previously unavailable to workers, especially those in remote or underserved areas (Zheng et al. 2022). By removing physical barriers to education and offering flexibility in terms of when and where learning takes place, digital learning democratizes access to knowledge, enabling workers to accumulate human capital regardless of their location or socio-economic background.

Therefore, our paper seeks to answer a key question: Can digital learning solve the problem of the ineffectiveness of traditional vocational training and achieve positive market returns?

First, we develop a novel theoretical model to explain what is the difference between the impact of digital learning and traditional vocational training. We develop an on-the-job training model adding online learning. We find that online learning can accumulate human capital and reduce learning costs to increase more income than offline learning. This conclusion coincides with the normal situations, regardless of who pays for vocational training fees.

Then, we examine the relationship between digital learning and labor market outcomes by empirical model. We demonstrate that digital learning has both short-and long-term income growth effects. The results suggest that digital learning can increase labor returns by approximately 6.9%. Moreover, we find that digital learning plays a crucial role in both other monetary labor market outcomes and non-monetary labor market outcomes. Our results are robust after employing the Bartik IV approach and propensity score matching (PSM).

In addition, we prove mechanisms through which digital learning influences labor market outcomes based on the theoretical model. The mechanism analysis reveals that digital learning operates through two main channels: first, by accumulating human capital, especially for workers with lower initial education levels, and second, by reducing the costs of learning, including both direct costs and opportunity costs.

Furthermore, we investigate which groups benefit the most from digital learning. Our results show that individuals aged over 35 or engaged in non-physical occupations experience significant positive effects from digital learning. However, we find no significant difference between males and females. These findings help to clarify the target groups for digital learning and understand the difference between digital learning and traditional offline learning.

Our study contributes primarily to three strands of literature. First, it adds to the literature on vocational training. The usefulness of the role of vocational training is still being debated. Some studies demonstrate its positive impact on income (Jacobson et al. 2005; Böckerman et al. 2019; Bratsberg et al. 2020; Takasaki 2024). However, some studies question the overall effectiveness of traditional vocational training. For instance, Schwerdt et al. (2012) found no significant labor market returns from adult education programs in Switzerland, and Hogendoorn et al. (2019) noted that vocational training had little effect on fostering innovation and entrepreneurship within firms.

These mixed findings may be partly attributed to the failure to choose who are the targets of vocational training. Existing positive evidence shows that vocational training can be particularly effective for disadvantaged groups (Abramovsky et al. 2011; Janzen et al. 2025; Mata et al. 2025), while others find little effect on highly skilled people (Schwerdt et al. 2012). But traditional training programs tend to favor highly skilled or highly educated individuals (Lynch 1992; Barron et al. 1999; Parent 1999; Acemoglu and Pischke 1999a, 1999b). So it is hard to find the effect of traditional training programs unless only targeting disadvantaged groups. However, digital learning reduces training costs, and some disadvantaged groups can participate in vocational training. We find that digital learning increases income, especially for people with low education and middle-aged and elderly people. Our findings offer a reasonable explanation for the existing debate on the role of vocational training.

Second, our paper also contributes to the literature on the effect of digital learning. There is no consensus on this topic either. Some studies highlight its positive impact on education outcomes (Tchamyou et al. 2019; Lu and Song 2020; Bianchi et al. 2022), while others express concerns about the effectiveness of online learning (Figlio et al. 2013; Novella et al. 2024). Some critics argue that the lack of accountability and motivation in digital learning environments can undermine its effectiveness (Novella et al. 2024). However, the above literature mainly studies the impact on students' digital learning, and few studies consider the impact of adult digital learning. Compared with students, adults have more initiative in learning and are easier to discover the essential role of digital learning itself. Therefore, we examine the positive effect of digital learning, specifically excluding the potential negative effects attributed to low motivation or a lack of incentives.

Third, our paper focuses on the relationship between digital learning and labor market performance. But the literature is limited. Most studies focus on early formal education (Bulman and Fairlie 2016; Tchamyou et al. 2019; Lechman and Popowska 2022; Meng et al. 2023; Lakdawala et al. 2023), with few examining the role of digital learning in vocational training. Additionally, studies like those by Lu and Song (2020) and Novella et al. (2024) address digital learning's impact in the formal education stage but do not explore its role in post-education workforce training. This paper fills that gap by examining the effects of adult digital learning on labor market returns.

The structure of the paper is as follows: Section 2 outlines the theoretical model, Section 3 introduces the empirical model and data sources, Section 4 presents the empirical analysis, Section 5 discusses external validity, and Section 6 concludes with policy recommendations.

2 | Theoretical Model

We adopt a simple model to explain the differences between online training and offline training, thereby providing theoretical support for our empirical results. Based on a previous on-the- job training model only considering offline training (Ben-Porath 1967; Jovanovic 1979; Heckman et al. 1998; Acemoglu and Pischke 1999a, 1999b; Falvey et al. 2010; Fudenberg and Rayo 2019; Ma et al. 2024), we add the online training.

The economy consists of infinitely lived consumers, final goods producers, and two types of training firms. Each consumer accumulates human capital by purchasing educational goods and obtains utility through consuming final goods and leisure time. We assume online and offline training are completely substitutable. That is, the problem of consumer utility maximization is as follows:

$$\max \sum_{t=0}^{\infty} \beta^t \left(lnC_t + \phi lnL_t \right) \tag{1}$$

where β is the discount rate for each period, C_t is the consumption in period t, and L_t is the leisure time in period t. Then, we assume that all time in each period for consumers is limited, and we normalize it to 1, namely:

$$L_t + N_t + e_{ft} + e_{ot} = 1 \tag{2}$$

where N_t is the labor time in period t, and e_{ft} and e_{ot} are the offline and online training time in period t, respectively.

In addition, consumers are confronted with budget constraints:

$$C_t + P_{ft}Q_{ft} + P_{ot}Q_{ot} = W_t H_t N_t \tag{3}$$

where P_{ft} and P_{ot} are the prices of offline training and online training in period t respectively, W_t is the salary in period t, and H_t is the human capital level in period t. Q_{ft} and Q_{ot} are the goods which offline training and online training use in period t respectively, which can be considered as the context of what learned.

Consumers accumulate human capital H through training each period:

$$H_{t+1} = (1-\delta)H_t + e_{ft}^{\alpha_f} Q_{ft}^{1-\alpha_f} + e_{ot}^{\alpha_o} Q_{ot}^{1-\alpha_o} \tag{4}$$

where α_f and α_o represent the degree which offline and online training rely on time investment, respectively. Specifically, we assume that offline training is more time-intensive, whereas online training relies more on platform-based delivery and digital content. Formally, we set $\alpha_f > \alpha_o$, indicating that online training is less dependent on direct time input than resource input.

First, we solve for the part equilibrium of maximizing consumer utility.[1] By Equation (4), we can easily know online training can accumulate human capital. But online training also reduces the labor time N_t, it is hard to see the effect on lifelong income. To simplify the model, we assume that $\phi = 0$. We show the changes in human capital and cumulative income of offline training and online training through numerical simulation. We assume $\beta = 0.96$, $\delta = 0.05$, $\alpha_f = 0.6$, $\alpha_o = 0.3$ and $H_0 = 0.5$ to solve the model given the parameters. Then, we show the results in Figure 1. As time goes by, online training offers more growth in human capital accumulation and cumulative income compared to offline training. Thus, we propose Hypothesis 1:

Hypothesis 1. *Online training increases income by accumulating human capital.*

Then, we conduct a comparative static analysis of the model. By maximizing consumer utility, we obtain:

$$\frac{e_{ft}^{1-\alpha_f}}{e_{ot}^{1-\alpha_o}} = \frac{\alpha_f Q_{ft}^{1-\alpha_f}}{\alpha_o Q_{ot}^{1-\alpha_o}} \tag{5}$$

We assume that the learning content online and offline is the same, namely $Q_{ft} = Q_{ot} = 1$. Combine with the Formula (5), we

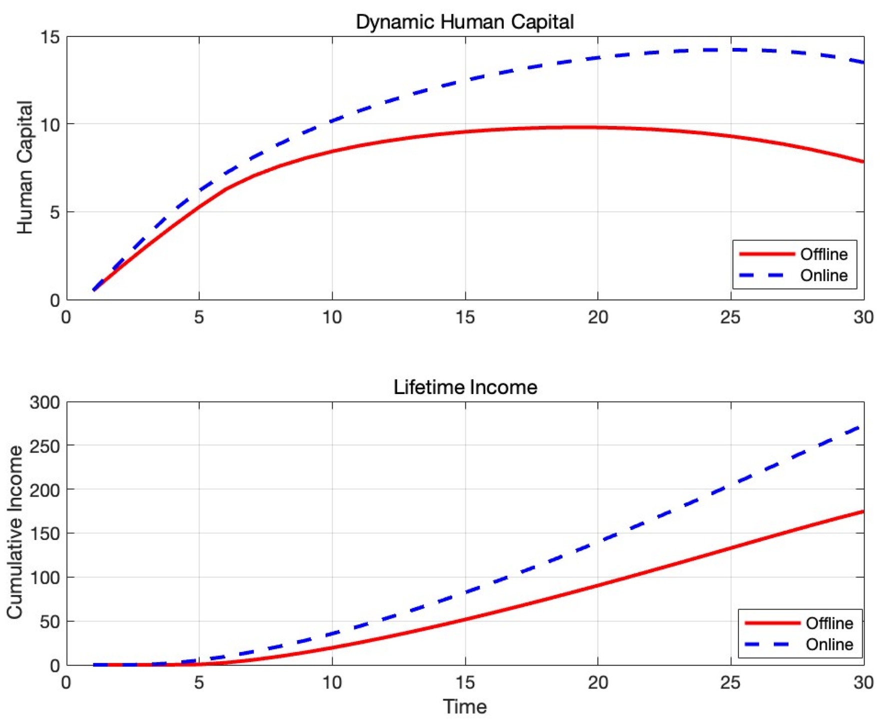

FIGURE 1 | The human capital and cumulative income of offline training and online training. The red solid line represents offline training, while the blue dash line represents online training. The figure above is the changes in human capital, while the figure below is the changes in lifetime income.

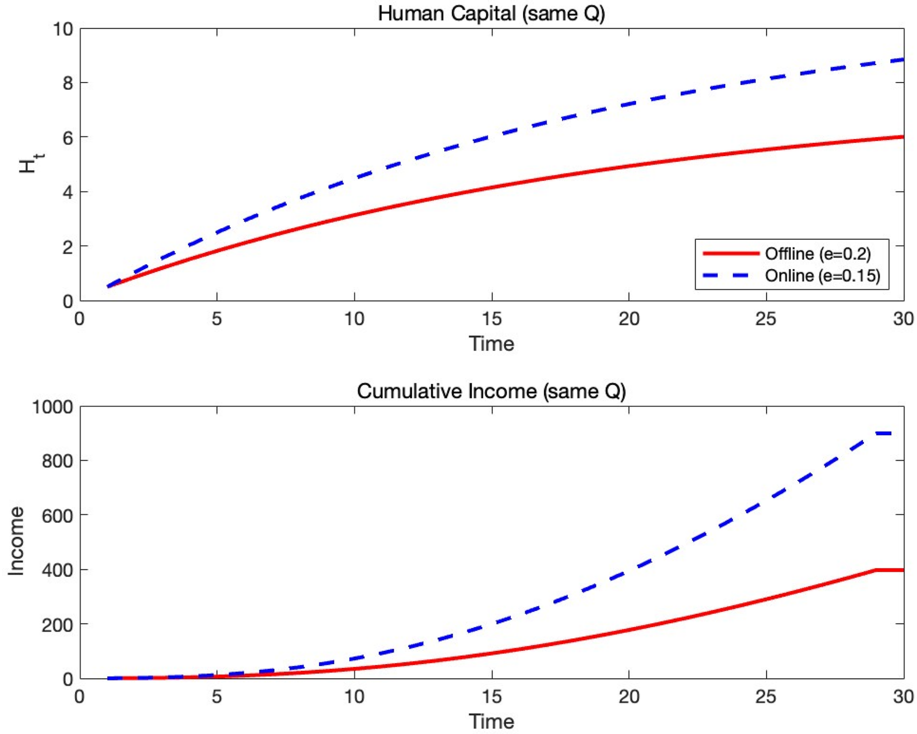

FIGURE 2 | The human capital and cumulative income of the same learning context. The red solid line represents offline training, while the blue dash line represents online training. The figure above is the changes in human capital, while the figure below is the changes in lifetime income.

know that online training uses less time to learn the same. We also simplify the model through numerical simulation as before as shown in Figure 2. We assume $e_{ft} = 0.2$ to calculate e_{ot}. When $Q_{ft} = Q_{ot} = 1$, online training uses less time than offline training, thereby increasing human capital and income. Therefore, we propose Hypothesis 2:

Hypothesis 2. *Online training increases income by reducing the time cost of training.*

In terms of firms, we have three types of firms: those that produce final goods, and offline and online training firms that produce educational products. For firms that produce final goods, we assume that they use labor as an input factor for production, that is

$$Y_t = Z_t H_t N_t \tag{6}$$

Among them, Z_t represents the overall productivity of the enterprise. For enterprises that produce final goods, they will maximize profits:

$$\max Y_t - W_t H_t N_t$$

Then we can obtain:

$$W_t = Z_t \tag{7}$$

For training companies, they use capital as an input factor for production; that is

$$Q_{ft} = A_{ft} K_{ft} \tag{8}$$

$$Q_{ot} = A_{ot} K_{ot} \tag{9}$$

Among them, A_{ft} and A_{ot} represent the effects of offline training and online training respectively, and K_{ft} and K_{ot} represent the capital required for offline training and online training respectively, with their prices being r_{ft} and r_{ot} respectively. So, their issues regarding maximizing profits are respectively:

$$\max P_{ft} Q_{ft} - r_{ft} K_{ft}$$

$$\max P_{ot} Q_{ot} - r_{ot} K_{ot}$$

Then we can obtain:

$$P_{ft} = \frac{r_{ft}}{A_{ft}} \tag{10}$$

$$P_{ot} = \frac{r_{ot}}{A_{ot}} \tag{11}$$

According to Formulas 10 and 11, we find that the cost of training is closely related to the capital used for training. Because offline training requires more investment in fixed assets, such as venues and teaching equipment, we assume $r_{ft} > r_{ot}$. Thus, we know that $P_{ft} > P_{ot}$, which is in line with reality. By maximizing consumer utility, we also obtain:

$$\frac{P_{ft}}{P_{ot}} = \frac{\frac{\left(\frac{1}{\alpha_f} - 1\right) e_{ft}}{Q_{ft}}}{\frac{\left(\frac{1}{\alpha_o} - 1\right) e_{ot}}{Q_{ot}}} \tag{12}$$

If $P_{ft} > P_{ot}$ and we assume the time investment is the same, namely $e_{ft} = e_{ot}$, then we can find $Q_{ft} < Q_{ot}$. It means that we can

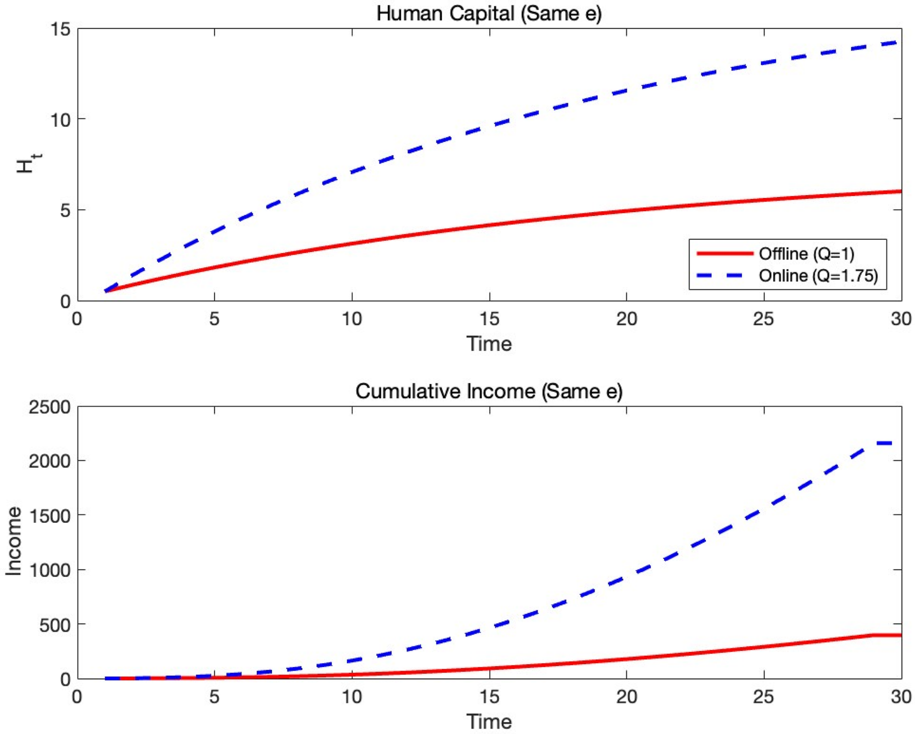

FIGURE 3 | The human capital and cumulative income of the same time investment. The red solid line represents offline training, while the blue dash line represents online training. The figure above is the changes in human capital, while the figure below is the changes in lifetime income.

learn more by online training than offline training with the same time investment. We assume $e_{ft} = e_{ot} = 0.2, A_{ft} = A_{ot} = 1, r_{ft} = 2, r_{ot} = 1$. Then we also simplify the model through numerical simulation as before as shown in Figure 3. We find when the time investment is the same, online training can learn more than offline training, thereby increasing human capital and lifelong income. Thus, the Hypothesis 3 can be derived:

Hypothesis 3. *Digital learning increases income by reducing the direct cost of learning.*

In conclusion, through a simple theoretical model, we have found that digital learning accumulates human capital by reducing the opportunity cost (time cost) and direct cost of learning, thereby increasing income. Moreover, these findings are in line with the actual situation. Even if the costs are borne by the enterprises, the conclusion still holds.[2] Therefore, digital learning maybe could replace offline training as the best choices of individuals and firms.

3 | Model, Data and Index

3.1 | Model

To estimate the impact of digital learning on labor income, we employ a fixed-effects panel regression model specified as follows:

$$Y_{it} = \beta_0 + \beta_1 \text{digital learning}_{it} + \mathbf{X_C} \boldsymbol{\beta_C} + \sigma_i + \lambda_t + u_{it} \tag{13}$$

where the subscripts i and t denote the individual and time, respectively. The dependent variable Y_{it} represents the individual's labor market outcomes, including monetary and non-monetary labor market outcomes. Particularly, we focus on labor income as a main measure of the return to digital learning, which reflects the return of digital learning.

The key explanatory variable, digital learning$_{it}$, is a binary indicator that takes the value of 1 if the individual participates in digital learning in year t, and 0 otherwise. The vector $\boldsymbol{X_C}$ includes a set of individual-level control variables such as age, age squared, marital status, education level, gender, hukou (household registration), ethnicity, and health status. Household-level characteristics, including family size and dependency ratio, are also controlled for. The model includes individual fixed effects, denoted by σ_i, and time fixed effects, λ_t. Standard errors are clustered at the household level to account for potential within-household correlation.

3.2 | Data

The data used in this study is primarily sourced from the China Family Panel Studies (CFPS), a comprehensive database that collects information at the individual, family, and community levels. It reflects changes in Chinese society, economy, population, education, and health (Xie and Hu 2014). The survey is conducted biennially, and since 2014, it has included questions regarding individual internet usage, which is crucial for identifying digital learning participation.

This study utilizes micro-data from the years 2014, 2016, 2018, 2020, and 2022. After excluding observations with missing values and samples who have retired or are still in school, the sample is

TABLE 1 | Summary statistics.

	N	Mean	SD	Min	Max
ln(Labor income)	52,194	9.543	1.380	0.693	11.9
Digital learning	52,194	0.232	0.422	0	1
Age	52,194	43.661	11.315	16	65
Age square	52,194	2034.292	981.065	2,56	4225
Martial status	52,194	0.883	0.321	0	1
Education level	52,194	8.321	4.571	0	22
Gender	52,194	0.530	0.499	0	1
Hukou	52,194	0.206	0.404	0	1
Ethnicity	52,194	0.912	0.283	0	1
Health status	52,194	2.883	1.182	1	5
Family size	52,194	4.314	1.953	1	21
Dependency ratio	52,194	0.196	0.181	0	1

restricted to workers aged 16–65, resulting in a final sample size of 52,194 observations. Descriptive statistics for the sample are provided in Table 1.

3.3 | Index

Our key explanatory variable, digital learning, is derived from survey questions on internet usage for educational purposes. In survey waves prior to 2020, respondents were asked: "How frequently do you use the internet for learning activities?" with seven ordered response categories: (1) almost daily, (2) 3–4 times/week, (3) 1–2 times/week, (4) 2–3 times/month, (5) once/month, (6) once every few months, and (7) never. We define digital learning as a binary indicator equal to 1 if the respondent reports any positive frequency (options 1–6), and 0 if they selected "Never" (option 7). In the 2020 and 2022 survey waves, the question wording changed to: "Have you used online learning in the past week?" A "Yes" response is coded as 1; otherwise, it is 0.

Although the problems in 2020 and 2022 changed, at most, our results are underestimated. First, the inquiries in 2020 and 2022 limited the time range of the previous week, but due to random surveys, they can reflect the individual Internet learning situation under normal circumstances. Second, as 2020 and 2022 limited the time range of last week, it might lead to some samples of Internet learning being considered to have no Internet learning. However, this would only underestimate our estimation results.

In all, we find approximately 23.2% of the samples have been digitally learned. This is consistent with the statistics in the 55th "Statistical Report on the Development of China's Internet" released by the China Internet Society (CNNIC). The report says number of online education users in China has reached 355 million until 2024, accounting for approximately 25% of the total population in China. It also proves the validity of our key variable.

Due to data constraints, there are certain limitations in identifying the effect of digital learning. First, the measures are based on self-reported information, which may be subject to misclassification or recall bias. However, our sample is sufficiently large to make occasional reporting errors unlikely to materially affect the results. Moreover, even if misclassification occurs—such as respondents who never engaged in digital learning mistakenly reporting participation, or actual participants reporting otherwise—this would most likely underestimate the effect. It implies that our results represent a conservative estimate. Second, there are data limitations regarding the more detailed information of learning, such as quality, content, and so on. We are unable to observe detailed learning modalities or arrangements. Only the 2022 wave includes information on the daily duration of online learning, and no survey wave provides data on who bears the cost of training.

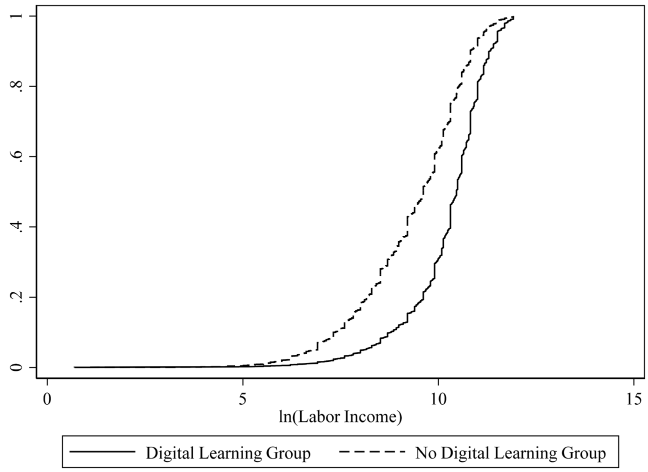

FIGURE 4 | Cumulative density distributions of labor income. The digital learning group is represented by a solid line, while the no digital learning group is represented by a dashed line.

In Figure 4, we plot the cumulative density distributions of labor income for the digital learning and no digital learning groups. The figure clearly illustrates that the income distribution for the digital learning group is significantly higher than that of the no digital learning group, offering preliminary support for Hypothesis 1. This suggests that digital learning is associated with higher income. Further empirical analysis will be conducted to validate this preliminary finding.

4 | Regression Results

4.1 | Baseline Results

To estimate the labor market returns of digital learning, we first examine both the short- and long-term relationships between digital learning and income. Additionally, we explore the impact of digital learning on other aspects, including monetary labor market outcomes, non-monetary labor market outcomes, and general skills.

4.1.1 | The Impact of Digital Learning on Income

Table 2 presents the results of the baseline regression analysis. Columns (1) through (4) progressively add control variables and fixed effects, and the results of full control variables are in

TABLE 2 | Baseline results.

Variables	(1)	(2)	(3)	(4)
	ln(Labor income)			
Digital learning	0.893***	0.239***	0.274***	0.069***
	(0.015)	(0.015)	(0.014)	(0.016)
Controls	No	Yes	Yes	Yes
Year fixed effect	No	No	Yes	Yes
Individual fixed effect	No	No	No	Yes
Constant	9.336***	7.355***	7.850***	6.761***
	(0.011)	(0.108)	(0.107)	(0.711)
Observations	52,194	52,194	52,194	52,194
Adjusted R-squared	0.075	0.259	0.292	0.580

Note: The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

Table A1. In all specifications, the coefficient for digital learning is significantly positive at the 1% level. Specifically, the results in column (4) demonstrate that after controlling for individual and time fixed effects, the coefficient of digital learning remains significantly positive at the 1% level. This provides strong evidence that participation in digital learning is associated with a substantial increase in income, offering support for Hypothesis 1. This finding aligns with existing literature, which reinforces the positive relationship between digital learning and income (Lu and Song 2020).

In terms of economic significance, participation in digital learning is associated with an approximate 6.9% increase in individual income. This result is economically significant when compared to existing literature. For instance, the return to higher education in China is approximately 40% (Li et al. 2025, 2012). This aligns with the general finding in the literature that the return on vocational training is typically lower than that of formal education. Additionally, Lu and Song (2020) reported that the labor market return of information and communications technology (ICT) for Chinese college students is 3.2%, less than half of our result. This comparison highlights that the impact of digital learning on income not only stems from the advantages of computer technology but also reflects the accumulation of human capital and the subsequent improvement in workers' market performance.

In summary, the results presented in Table 2 confirm that digital learning significantly increases income and enhances labor market returns for adults. By utilizing online platforms, digital learning enables workers to efficiently accumulate human capital during their leisure time, thereby boosting their labor market performance.

4.1.2 | The Long-Term Dynamic Impact of Digital Learning on Income

To further investigate the long-term dynamic effects of digital learning, we employ an event study methodology. The regression equation for the event study is specified as follows:

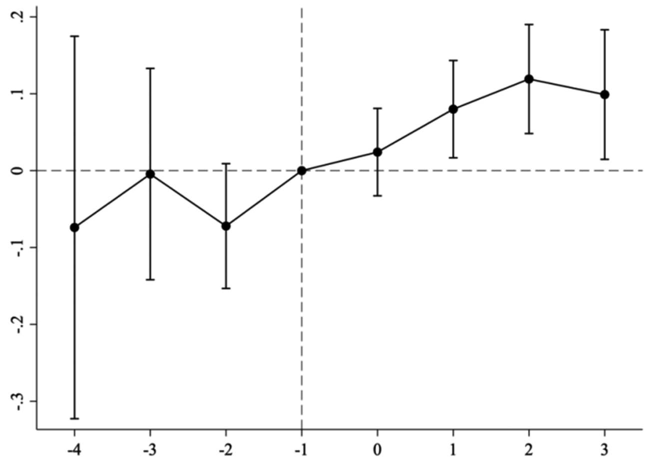

FIGURE 5 | The long-term dynamic impact of digital learning on income. The horizontal axis represents the relative time of an individual's first engagement with digital learning, while the vertical axis shows the estimated coefficient size of the dynamic effects. The black dots represent the coefficient estimates of the dynamic effects. The solid black lines represent the 95% confidence intervals.

$$\ln(\text{income})_{it} = \beta_0 + \sum_{j\neq-1} \beta_j 1\left\{l = t - G_i\right\} + \mathbf{X}_C \boldsymbol{\beta}_C + \sigma_i + \lambda_t + u_{it} \tag{14}$$

In this equation, $1\left\{l = t - G_i\right\}$ is a dummy variable that indicates the time relative to an individual's first engagement in digital learning. The period prior to the first digital learning experience is used as the baseline, and other specifications are consistent with the baseline regression. The results are presented in Figure 5.

First, we check the parallel trend test. Prior to the event, the estimated coefficients for the pre-treatment periods (to the left of the dashed line) are close to zero. Their 95% confidence intervals include zero, indicating no significant pre-trend differences between the treatment and control groups. Furthermore, we conduct a joint significance test (F-test) for all pre-treatment coefficients and fail to reject the null hypothesis that they are jointly zero (p value = 0.345), providing statistical support for the parallel trends assumption.

However, after the event, we observe a significant positive effect with a clear upward trend. This suggests that active participation in digital learning not only contributes to the accumulation of human capital but also leads to a sustained, positive long-term impact on income.

To further assess the robustness of our results and rule out the possibility that the estimated effects are driven by spurious correlations or unobserved shocks, we conduct a placebo test. For each year, we randomly assign the treatment status to a group of individuals who were not actually treated and re-estimate the event study specification using this falsified treatment assignment. This procedure is repeated 500 times to generate the empirical distribution of placebo estimates. Figure 6 presents the distribution of these simulated coefficients of aggregate effect. The solid vertical line indicates the actual estimated aggregate treatment effect from the event study analysis. As shown, the true estimate lies far in the tail of the placebo distribution, suggesting that the observed

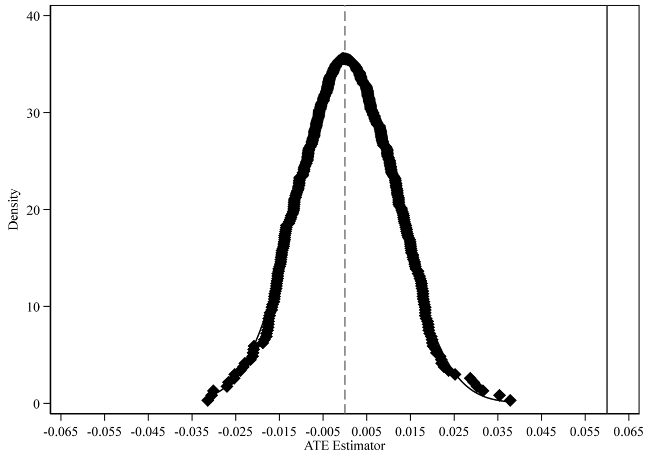

FIGURE 6 | Placebo test. The solid line is the result of five hundred simulations, and the dashed line is represents the aggregate treatment effect estimated by the event study model.

effect is unlikely to be driven by random variation. This reinforces the credibility of our identification strategy.

4.1.3 | The Impact of Digital Learning on Workers' Performance

We have established the positive impact of active digital learning on income, but this analysis has been limited to individuals with income. To further validate the effects of digital learning, we expand the effect to include a broader range of other workers' performance, including monetary labor market outcomes, non-monetary labor market outcomes, and general skills.

Firstly, we test the impact of digital learning on other monetary labor market outcomes. Specifically, we examine its impact on non-farm employment among rural populations, full-time employment, part-time employment, and primary job income. The results are presented in Panel A in Table 3. In column (1), we find that digital learning significantly promotes non-farm employment at the 1% level. Specifically, it increases the probability of non-farm employment for rural people by 3.4%. Next, we investigate the broader labor market performance of the entire workforce sample, focusing on whether digital learning aids in promoting full-time employment or part-time work. The results in columns (2) and (3) show that digital learning increases the likelihood of securing a job and obtaining part-time employment by approximately 2.5% and 3.2%, respectively. This indicates that digital learning not only boosts employment prospects but also helps adults diversify their labor market engagement. Furthermore, we exclude self-employed individuals from the sample and examine the effect of digital learning on primary job income. The results in column (4) show that digital learning has a significant positive effect on primary job income, with the coefficient statistically significant at the 1% level.

Secondly, we focus on non-monetary labor market outcomes, particularly individuals' job satisfaction. We measure satisfaction across multiple dimensions, including overall job satisfaction, job security, work environment, working hours, and opportunities for promotion.[3] As shown in Panel B, digital learning significantly enhances individuals' overall job satisfaction, with positive effects observed across various dimensions such as job security, work environment, working hours, and promotion opportunities. These findings suggest that the benefits of digital learning extend beyond income gains, contributing to job quality and career development. Besides, job satisfaction is linked to higher productivity and lower turnover (Antecol and Cobb-Clark 2009; Clark and Oswald 1996; Gao and Smyth 2010). Moreover, this provides policy suggestions for the transformation of developing countries and the retention of talents by enterprises in developed countries.

Thirdly, we investigate the effect on general skills. The data of CFPS have conducted literacy tests on respondents in 2014, 2018, and 2022. The literacy test is designed based on the Guttman Scale in psychometrics. The interviewer shows the respondents 34 Chinese characters in ascending order of difficulty. If the respondents answer three questions wrongly in a row, the test is terminated. The sequence position of the most difficult Chinese characters correctly answered by the respondents determines their scores. We take the score of the vocabulary test as the proxy variable for general skills and use this as the explained variable for regression. The results are shown in Panel C in Table 3. Individuals improve their vocabulary test scores through digital learning, which indicates that digital learning provides training in general knowledge. It differs from traditional vocational training focusing on special skills (Becker 1983).

In conclusion, our findings provide strong evidence that digital learning enhances adult workers' performance in the labor market, leading to improved labor market returns across multiple dimensions.

4.2 | Robustness Check

In the previous section, we demonstrate that digital learning significantly improves adult income and enhances labor market returns. However, several potential concerns remain. First, we examine the robustness of our results to alternative measurements of the key explanatory variable by replacing the binary indicator with the frequency of digital learning participation. Second, our estimates may be subject to omitted variable bias. The observed average treatment effect of digital learning might be influenced by unobserved factors, such as motivation or ability. To address this concern, we implement an instrumental variable (IV) approach and some matching methods to balance observable characteristics between treated and untreated groups. These additional analyses reinforce the robustness and credibility of our main findings.

4.2.1 | Replace Explanatory Variable

To test the robustness of our findings, we re-estimate the model using alternative measures of digital learning intensity. The results are presented in Table 4. We adopt three different approaches to measure the extent of digital learning. First, we construct a frequency-based index using the original survey responses, where the answers "Never," "Once every few months," "Once/month," "2–3 times/month," "1–2 times/week,"

TABLE 3 | The impact of digital learning on workers' performance.

Panel A: The effect on monetary labor market outcomes

Variables	(1) Rural Non-farm employment	(2) All Employment	(3) All Part-time employment	(4) Non-self-employed ln(Main income)
Digital learning	0.034***	0.025***	0.032***	0.053***
	(0.006)	(0.003)	(0.006)	(0.016)
Controls	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes	Yes
Constant	−0.065	0.339***	0.038	6.428***
	(0.179)	(0.126)	(0.202)	(0.632)
Observations	57,058	80,478	72,781	30,074
Adjusted R-squared	0.689	0.463	0.144	0.509

Panel B: The effect on non-monetary labor market outcomes

Variables	(1) Overall	(2) Safety	(3) Environment	(4) Working time	(5) Promotion
Digital learning	0.037***	0.040**	0.044**	0.049**	0.076***
	(0.014)	(0.018)	(0.018)	(0.019)	(0.024)
Controls	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes	Yes	Yes
Constant	3.145***	1.988***	3.437***	3.673***	1.803
	(0.529)	(0.500)	(0.742)	(0.709)	(2.092)
Observations	52,096	37,774	37,773	37,774	26,401
Adjusted R-squared	0.296	0.311	0.329	0.314	0.265

Panel C: The effect on common skills

Variables	(1) Scores for words test
Digital learning	0.819***
	(0.249)
Controls	Yes
Year fixed effect	Yes
Individual fixed effect	Yes
Constant	47.142***
	(10.228)
Observations	23,601
Adjusted R-squared	0.673

Note: All specifications control for year fixed effects and individual fixed effects. The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

TABLE 4 | Robustness check.

Variables	(1) ln(Labor income)	(2) ln(Labor income)	(3) ln(Labor income)
Digital learning frequency	0.009***		
	(0.003)		
ln(Digital learning times + 1)		0.009***	
		(0.003)	
ln(Digital learning duration + 1)			0.054***
			(0.018)
Controls	Yes	Yes	No
Year fixed effect	Yes	Yes	No
Individual fixed effect	Yes	Yes	No
Community fixed effect	No	No	Yes
Constant	6.695***	6.691***	8.083***
	(0.750)	(0.749)	(0.402)
Observations	51,817	51,817	4899
Adjusted R-squared	0.581	0.581	0.367

Note: All specifications control for year fixed effects and individual fixed effects. The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

"3–4 times/week," and "Almost daily" are coded from 0 to 6, respectively. Second, we convert the reported frequency into an estimated annual number of online learning sessions. Third, we use the reported average daily duration of online learning in 2022 as the explanatory variable.

As shown in Table 4, all results are statistically significant at the 1% level and show a positive effect. Specifically, Column (1) indicates that each 1% increase in the digital learning frequency index is associated with a 0.9% increase in income. Given that the average value in the treatment group is 4.4, this implies that digital learning raises income by approximately 4.0%. Furthermore, a 1% increase in the annual frequency of digital learning and in average daily digital learning time corresponds to a 0.9% and 5.4% increase in labor income, respectively. Similarly, Columns (2) and (3), with treatment group means of 4.6 and 1.7, suggest income increases of 4.1% and 9.0%, respectively. These findings are consistent with our baseline estimate. The slightly smaller effects in Columns (1) and (2) likely reflect variation within the treatment group, capturing differences in learning frequency beyond the treatment–control comparison. Additionally, the larger effect in Column (3) may be due to data limitations: since this measure is only available for 2022, we cannot control for individual or time fixed effects and instead rely solely on community fixed effects, which may inflate the estimated coefficient.

In all, this suggests that the positive effects of digital learning are robust even after replacing the explanatory variable.

4.2.2 | Instrumental Variable Method

To address potential endogeneity concerns stemming from omitted variables, we adopt an instrumental variables (IV) approach. Specifically, we construct a Bartik-style shift-share instrument that exploits plausibly exogenous variation induced by the *Broadband China* policy, a national policy launched in 2014 to expand broadband infrastructure across Chinese regions (Chen 2025). Since digital infrastructure is a prerequisite for engaging in digital learning, the rollout of this policy provides a natural source of variation in individuals' exposure to digital learning opportunities.

Following the methodology in Goldsmith-Pinkham et al. (2020), our instrument is defined as the interaction between community-industry employment shares and subsequent changes in digital learning rates at the community-industry level:

$$\text{Bartik IV}_{ict} = \sum_{k} \text{Share}_{ickt} \times \text{Shift}_{ckt}$$

where i indexes individual, c denotes community, k denotes industry, and t indicates the year. The exposure share Share_{ickt} is defined as the share of workers in industry k in community c in year t:

$$\text{Share}_{ickt} = \frac{\text{people}_{ckt}}{\sum_k \text{people}_{ckt}}$$

The shift component Shift_{ckt} captures the relative change in the proportion of workers engaging in digital learning in each community-industry cell over a two-year horizon[4]:

$$\text{Shift}_{ckt} = \frac{\text{rate}_{ck,t} - \text{rate}_{ck,t-2}}{\text{rate}_{ck,t-2}}$$

where $\text{rate}_{ck,t}$ is the proportion of workers engaging in digital learning in industry k in community c in year t. In all, we use this Bartik IV captures the exogenous change in digital learning because of the exogenous policy.

The regression results are presented in Columns (1) and (2) in Table 5. In Column (1), it shows that the instrument is relevant, with the coefficient for the instrument significantly positive at the 1% level. The Cragg-Donald Wald F-statistic and Kleibergen-Paap rk Wald F statistic both are very big, confirming that the instrument passes the weak instrument test.

In column (2), we find digital learning increases the income at the 5% significant level. It shows that workers participating in digital learning can increase their income by 31.1%, which is about four times larger than our baseline result. It is because the estimation of IV results is the local average treatment effect for individuals whose digital learning decisions are affected by exogenous variation in broadband infrastructure.

In all, after using the IV method to solve omitted problems, our coefficients are still significantly positive, validating the results.

TABLE 5 | Address endogeneity problems.

Variables	(1) First stage of IV Digital learning	(2) Second stage of IV ln(Labor income)	(3) PSM ln(Labor income)
Bartik IV	0.078***		
	(0.004)		
Digital learning		0.311**	0.068***
		(0.143)	(0.016)
Controls	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes
Constant	1.422***	6.418***	6.848***
	(0.228)	(0.746)	(0.709)
Observations	52,194	52,194	46,059
Cragg-Donald Wald F statistic	884.487		
Kleibergen-Paap rk Wald F statistic	327.474		
Adjusted R-squared	0.692	0.750	0.556

Note: All regressions control for year fixed effects and individual fixed effects. The standard errors are clustered at the household level. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

4.2.3 | Propensity Score Matching

To address self-selection bias arising from individuals' subjective decisions to engage in digital learning, we employ the PSM method. Specifically, we implement 1:1 nearest-neighbor matching based on a set of observable control variables. The balancing test results, presented in Figure A1, indicate that after matching, the covariates are well balanced between the treatment and control groups. This suggests that the matched sample is comparable and supports the credibility of the PSM estimates.

Column (3) of Table 5 reports the estimated effect of digital learning on income after matching. The coefficient remains significantly positive at the 1% level and is closely aligned with the baseline estimate, indicating that the positive association is robust to the solution of self-selection.

To further validate our findings, we also apply Coarsened Exact Matching (CEM) as an alternative approach. After using the CEM method, the L1 imbalance statistic decreases markedly from 0.823 to 0.367, reflecting improved covariate balance.[5] The estimation results, reported in Table A3, remain statistically significant and positive and still close to the baseline estimator, reinforcing the robustness of the digital learning effect.

Together, these results suggest that self-selection based on observable characteristics is unlikely to fully account for the observed income gains associated with digital learning.

Besides the above, we also do other robustness checks in section Other Robustness Checks in Appendix, including adjusting clustering levels, modifying model specifications, excluding data from periods influenced by the pandemic, and incorporating interaction fixed effects. In all, we do many robustness checks and address endogeneity problems to validate our results.

4.3 | Mechanism Analysis

This section empirically investigates the mechanisms through which digital learning contributes to improved labor market returns, as outlined in the theoretical model of our study. Specifically, we focus on two primary channels: the accumulation of human capital and the reduction of learning costs.

Digital learning allows individuals to supplement formal education by enhancing their skills, which directly contributes to better labor market outcomes. Unlike traditional education, which is often limited in scope and duration, digital learning provides a flexible, ongoing means of skill development. This flexibility enables workers to continuously update and refine their abilities, making them more competitive and productive in the labor market.

4.3.1 | Accumulating Human Capital

First, we explore the role of human capital accumulation as a key driver of the observed increases in income. Obviously, on-the-job training can accumulate human capital, and the accumulation of human capital will promote income growth. However, compared with formal education, traditional offline on-the-job training has not found consistent evidence of an increase in income (Jacobson et al. 2005; Schwerdt et al. 2012; Armona et al. 2018; Hogendoorn et al. 2019; Böckerman et al. 2019; Bratsberg et al. 2020). This is because in the past, training was often targeted at people with high education (Lynch 1992; Barron et al. 1999; Parent 1999; Acemoglu and Pischke 1999a, 1999b), and the marginal benefit of training for their human capital was limited (Schwerdt et al. 2012). That's why it is difficult to discover the positive effect of on-the-job training. It is consistent with positive evidence, which has found that public training programs specifically designed for vulnerable groups have a significant effect (Abramovsky et al. 2011; Janzen et al. 2025; Mata et al. 2025). Based on this, we study whether digital learning accumulates human capital for people with low education.

We divide the samples for regression based on whether individuals have received higher education in column (1) and (2) in Table 6. The results reveal that digital learning has a statistically significant positive effect on the income of individuals without higher education, whereas the impact is not significant for those with higher education. This finding suggests that digital learning and formal education are complementary, with digital learning helping to fill the gaps left by formal education, thereby facilitating the accumulation of human capital.

The result contrasts with much of the traditional training literature, which finds that high-skilled individuals tend to benefit more from firm-provided training (Jacobson and Davis 2017). It is either because firms preferentially invest in higher-skilled workers or because these individuals are more likely to be employed in large firms capable of bearing the cost of training (Lynch 1992; Barron et al. 1999; Parent 1999; Acemoglu and Pischke 1999a, 1999b). However, our finding aligns with recent research suggesting that when training is provided outside of firm discretion, such as through public policy, low-skilled individuals may benefit more (Schwerdt et al. 2012; Das 2021; Baird et al. 2024). In this context, digital learning appears to broaden access to human capital accumulation, alleviating disparities historically associated with education and firm-sponsored training.

In sum, our findings demonstrate that digital learning can accumulate human capital for people with low education. This highlights the importance of digital learning as a vital tool for enhancing labor market outcomes, particularly in an increasingly knowledge-driven economy.

TABLE 6 | Mechanism analysis.

Variables	(1) No higher education ln(Labor income)	(2) Higher education ln(Labor income)	(3) Short leisure time ln(Labor income)	(4) Long leisure time ln(Labor income)
Digital learning	0.083***	0.027	0.076***	0.042
	(0.019)	(0.032)	(0.029)	(0.028)
Controls	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes	Yes
Constant	6.687***	4.392***	8.494***	4.706***
	(0.696)	(1.251)	(0.952)	(1.529)
Observations	45,214	6980	30,752	20,882
Adjusted R-squared	0.547	0.417	0.582	0.597

Note: All regressions control for year fixed effects and individual fixed effects. The standard errors are clustered at the household level. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

4.3.2 | Reducing Training Costs

But why can people with low education benefit from digital learning to accumulate human capital? As shown in the theoretical model, that's because digital learning reduces the cost of on-the-job training. Second, we examine how digital learning reduces the costs traditionally associated with acquiring new skills. Vocational training, typically pursued after entering the labor market, is subject to considerable time constraints and opportunity costs (Armona et al. 2018). Traditional vocational training often demands long periods of offline instruction, during which workers must forgo income-generating activities. In contrast, digital learning, facilitated by online platforms, enables fragmented learning that can take place at any time and from any location, substantially reducing the costs associated with skill acquisition (Zheng et al. 2022).

To empirically test this hypothesis, we perform subgroup regressions that distinguish workers based on their available leisure time. We define leisure time as the sum of weekly exercise time, internet usage, and TV/movie watching time. Workers are then grouped based on the median value of leisure time. The results, presented in columns (3) and (4) of Table 6, reveal a significant positive impact of digital learning on the income of workers with lower leisure time, while no significant effect is observed for workers with higher leisure time. This finding supports our hypothesis that digital learning effectively reduces the costs of learning by leveraging flexible, fragmented learning schedules. In turn, this reduction in costs contributes to improved labor market returns, especially for people with low education.

4.4 | Heterogeneity Analysis

In this section, we investigate which groups derive the greatest benefits from digital learning by conducting subgroup analyses based on gender, age, and occupation as shown in Table 7.

TABLE 7 | Heterogeneous analysis.

	(1) Male	(2) Female	(3) Age < 35	(4) Age ≥ 35	(5) Non-physical	(6) Physical
Variables	ln(Labor income)					
Digital learning	0.064***	0.084***	0.043	0.079***	0.063***	0.069
	(0.021)	(0.025)	(0.027)	(0.022)	(0.017)	(0.058)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Constant	7.854***	4.808***	2.454	7.906***	7.926***	6.005***
	(0.786)	(1.076)	(1.555)	(1.171)	(0.854)	(1.840)
Observations	27,666	24,528	14,830	37,364	33,614	18,580
Adjusted R-squared	0.524	0.598	0.489	0.586	0.589	0.518

Note: All regressions control for year fixed effects and individual fixed effects. The standard errors are clustered at the household level. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

First, Columns (1) and (2) show no statistically significant difference between men and women. While the coefficient for women is marginally larger than that for men, the difference is not statistically significant. It suggests that digital learning benefits both genders in a broadly similar manner. But there are gender differences in offline training, and men tend to benefit more (Jacobson and Davis 2017; Brixiová et al. 2020). This is because offline training often requires a significant investment of time and money, which women are less able to commit to due to childcare responsibilities. Therefore, digital learning provides a feasible channel for women to participate in training during their spare time.

Second, we divide the sample at the age threshold of 35, reflecting the fact that many Chinese firms impose an implicit “35-year-old ceiling” in recruitment, which creates systematic disadvantages for older workers (Fang and Qiu 2023). Our results suggest that digital learning can help attenuate such age-based discrimination by enabling individuals aged above 35 to enhance their productivity and earnings through flexible, skill-oriented training.

Third, we classify occupations into physical and nonphysical labor. We find that income gains from digital learning are concentrated among nonphysical workers, consistent with the view that digital learning primarily fosters human capital accumulation relevant to knowledge-intensive and skill-based tasks. In contrast, physical labor occupations—often less compatible with digital skill application—show weaker returns.

Furthermore, we compare digital learning with offline training in section Other Heterogeneity Analysis in the Appendix. In contrast to traditional education, digital learning is not bound by geographical, temporal, or resource constraints, making it particularly advantageous for workers facing mobility barriers or time limitations.

Taken together, our findings indicate that, unlike offline training, digital learning has the potential to break implicit gender and age barriers and improve earnings in nonphysical occupations.

From a policy perspective, these results suggest that targeted promotion of digital learning platforms for older workers and those in knowledge-intensive roles could yield substantial labor market benefits. Subsidizing digital course fees, providing workplace-based digital learning incentives, and integrating digital skill certification into public employment services may help maximize these gains, particularly in contexts where age discrimination and occupational skill gaps remain persistent.

5 | Discussion on External Validity

Digital learning has important policy implications for on-the-job training and human capital accumulation, particularly in developing countries. Historically, developed economies have outperformed developing countries in the accumulation of human capital through on-the-job training, contributing directly to the income gap between them (Ma et al. 2024). In developed countries, firms with higher productivity can afford the cost of employee training, whereas firms in developing countries often rely on low labor costs from low-skilled workers to generate profits and are less willing to invest in training. Our subgroup regressions by income level and quantile regressions (Table A6) show that low-income individuals benefit more from digital learning than high-income individuals. This indicates that digital learning reduces the cost and entry barriers of vocational training, providing an effective pathway for those who previously lacked training opportunities. This finding offers a valuable reference for policymakers in developing countries: expanding training coverage through digital means.

Digital learning also helps alleviate the educational entry barriers present in traditional vocational training. Many studies have found substantial educational heterogeneity in the returns to training (Jacobson and Davis 2017). Traditional firm-sponsored training programs tend to favor highly skilled or highly educated individuals, either because firms prioritize training for high performers or because high-skilled workers are more likely to be employed in large firms that can afford such programs (Lynch 1992; Barron et al. 1999; Parent 1999; Acemoglu and Pischke 1999a). In our mechanism analysis, we find that digital learning has a significant effect on low-education individuals, compensating for the structural bias in which firms predominantly invest in high-skilled employees. This aligns with existing evidence that vocational training can be particularly effective for disadvantaged groups (Abramovsky et al. 2011; Janzen et al. 2025; Mata et al. 2025).

Moreover, digital learning is not unique to China but has rapidly expanded worldwide. For example, massive open online courses (MOOCs) have emerged globally since 2012, with over 220 million registered users on major platforms such as Coursera, edX, and Udacity by 2024. Among them, Coursera, the largest online education platform, received 1.5 million online learning registration applications within more than two months of its official launch. The number of online learners who successfully registered reached 640,000, coming from over 190 countries and regions. In February 2024, the Indian government, in collaboration with industry partners such as L&T, Microsoft, and Cisco, launched the “Swayam Plus” online vocational training platform to enhance employability and professional skills. Across Africa, countries are actively promoting digital education and implementing remote learning, laying the groundwork for the adoption of digital approaches in vocational training by future training institutions, governments, and enterprises.

Finally, digital learning is a critical tool for achieving lifelong learning. While community-based learning existed in developed countries in earlier decades, it was constrained by time, location, and cost. In contrast, digital learning offers greater accessibility and flexibility in any region (Zheng et al. 2022), making it particularly suitable in the context of global population aging and rapid artificial intelligence adoption. It enables workers to continually update their skills in response to technological shocks (Autor 2022) and lowers barriers to retraining for middle-aged and older adults, thereby improving the adaptability and resilience of labor markets worldwide (Mata et al. 2025). For instance, the median age of users studying on the Coursera platform in 2024 was 35 years old. This indicates that an increasing number of middle-aged and elderly people are beginning to use the Internet for learning. Besides, the United States pioneered the GetSetUp platform for older learners, which has provided services to over 4.4 million seniors in more than 160 countries. This is consistent with our heterogeneity analysis, which finds that digital learning more effectively boosts the income of middle-aged and older individuals, making it relevant for countries facing demographic aging.

Therefore, our study demonstrates strong external validity. Although our empirical analysis is based on Chinese data, China is both the largest developing country and the country with the largest number of digital learning users. Evidence from China can thus provide valuable insights for the development of digital learning in other contexts, particularly for developing economies and countries facing population aging.

6 | Conclusions

By combining theoretical and empirical analysis, we find that engagement in digital learning significantly boosts labor market returns, including both monetary and non-monetary outcomes. This result contrasts with the literature emphasizing potential negative impacts of digital learning (Figlio et al. 2013; Novella et al. 2024), which are often attributed to students' lack of learning motivation. Our study focuses on adults rather than students. Adults generally exhibit greater self-discipline, which allows digital learning to play a more effective role in vocational training. Therefore, the prerequisite for giving full play to the role of digital learning is to bring into play people's subjective initiative. This also explains the reasons for the differences in the effects of existing literature on digital learning (Figlio et al. 2013; Tchamyou et al. 2019; Lu and Song 2020; Bianchi et al. 2022; Novella et al. 2024).

The underlying mechanism is that digital learning simultaneously facilitates human capital accumulation and reduces the cost of training. This expands access to vocational training for low-income and low-skilled individuals, lowering entry barriers and enabling disadvantaged groups, latecomer firms, and developing economies to improve competitiveness through on-the-job training. This also explains why general vocational training is useless but vocational training for disadvantaged groups is effective (Abramovsky et al. 2011; Schwerdt et al. 2012; Janzen et al. 2025; Mata et al. 2025).

Our heterogeneity analysis identifies groups that benefit most from digital learning: middle-aged and elderly workers and non-manual laborers. And we also find no gender difference. This provides evidence-based recommendations for promoting lifelong learning and addressing the employment challenges of aging societies.

Given these findings, we propose the following policy implications. First, policy efforts should focus on promoting and integrating digital learning into national vocational training systems, supported by both government and enterprises. Expanding access to high-quality online courses, coupled with initiatives to enhance learner motivation, can amplify the benefits for disadvantaged groups, such as certification incentives, employer recognition, and career progression opportunities.

Second, targeted lifelong learning programs for middle-aged and elderly workers and non-manual laborers can help mitigate the employment challenges of aging societies. Thus, policymakers could consider providing targeted subsidies, workplace incentives, or community-based programs to encourage more frequent and longer-term engagement in digital learning, particularly for disadvantaged groups who may face higher barriers to participation.

Third, strengthening digital infrastructure and reducing connectivity costs will not only ensure equitable access within a country but also create positive externalities by fostering a more skilled and adaptable workforce, enhancing labor market efficiency, and improving overall economic resilience. Our results thus offer a scalable policy reference for leveraging digital learning to promote inclusive growth and global human capital development.

Our study is limited by the data, which do not distinguish between different digital learning platforms, content quality, or accreditation status. In particular, digital training accompanied by formal certification or diplomas is often more meaningful for labor market outcomes, as it signals verified skills to employers (Athey and Palikot 2024). Unfortunately, such distinctions are not available in our dataset, so our estimates reflect average effects across heterogeneous types of digital learning. Future research with richer data on certification, platform differences, and training intensity would provide deeper insights into the quality dimension and its impact on employment and income.

Ethics Statement

I certify that this manuscript is original and has not been published and will not be submitted elsewhere for publication while being considered by *Review of Income and Wealth*. And the study is not split up into several parts to increase the quantity of submissions and submitted to various journals or to one journal over time. No data have been fabricated or manipulated (including images) to support your conclusions. No data, text, or theories by others are presented as if they were our own. The submission has been received explicitly from all co-authors. And authors whose names appear on the submission have contributed sufficiently to the scientific work and therefore share collective responsibility and accountability for the results.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from China Family Panel Studies. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of China Family Panel Studies.

Endnotes

[1] The complete derivation process is in section The Full Deviation of Model in the appendix.

[2] The proof is in section The Full Deviation of Model in Appendix.

[3] This is achieved by asking individuals about their job satisfaction. The respondents choose a score from 1 to 5 and rate it.

[4] The two-year horizon is due to the survey design of data, which collects information at two-year intervals. And we set the value of shift in 2014 is 1.

[5] The L1 statistic measures the imbalance between the treatment and control groups, taking values between 0 and 1. Smaller values indicate better covariate balance.

[6] The dependent variable is based on the CFPS survey question, "How important is the internet in acquiring information?" The values range from 1 to 5, with higher values indicating greater importance.

[7] "guanxi income" refers to the income from offline interpersonal relationships.

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Appendix

The Full Deviation of Model

1. Consumers maximize their utility

$$\max \sum_{t=0}^{\infty} \beta^t \left(lnC_t + \phi lnL_t \right)$$

s.t.

$$C_t + P_{ft} Q_{ft} + P_{ot} Q_{ot} = W_t H_t N_t$$

$$H_{t+1} = (1-\delta) H_t + e_{ft}^{\alpha_f} Q_{ft}^{1-\alpha_f} + e_{ot}^{\alpha_o} Q_{ot}^{1-\alpha_o}$$

$$L_t + N_t + e_{ft} + e_{ot} = 1$$

We use Lagrange Multiplier Method to solve this problem:

$$\begin{aligned} \max\ L = \sum_{t=0}^{\infty} \beta^t \Big\{ & \left[\ln\left(W_t H_t N_t - P_{ft} Q_{ft} - P_{ot} Q_{ot} \right) + \phi \ln\left(1 - N_t - e_{ft} - e_{ot} \right) \right] \\ & + \lambda_t \left(H_{t+1} - (1-\delta) H_t - e_{ft}^{\alpha_f} Q_{ft}^{1-\alpha_f} - e_{ot}^{\alpha_o} Q_{ot}^{1-\alpha_o} \right) \Big\} \end{aligned}$$

Then, we can deviate from the first-order condition:

$$\frac{1}{C_t} = \frac{\phi}{L_t} \tag{A1}$$

$$\frac{e_{ft}^{1-\alpha_f}}{e_{ot}^{1-\alpha_o}} = \frac{\alpha_f Q_{ft}^{1-\alpha_f}}{\alpha_o Q_{ot}^{1-\alpha_o}} \tag{A2}$$

$$\frac{P_{ft}}{P_{ot}} = \frac{\left(1-\alpha_f\right) e_{ft}^{\alpha_f}\ Q_{ft}^{-\alpha_f}}{\left(1-\alpha_o\right) e_{ot}^{\alpha_o}\ Q_{ot}^{-\alpha_o}} \tag{A3}$$

$$\frac{\beta W_{t+1} N_{t+1}}{C_{t+1}} = \beta \lambda_{t+1} (1-\delta) - \lambda_t \tag{A4}$$

Combine (A2) and (A3), we can obtain:

$$\frac{P_{ft}}{P_{ot}} = \frac{\frac{\left(\frac{1}{\alpha_f} - 1\right) e_{ft}}{Q_{ft}}}{\frac{\left(\frac{1}{\alpha_o} - 1\right) e_{ot}}{Q_{ot}}}$$

2. The enterprise bears the training expenses

We consider the situation, which enterprise bears the training expenses. For firms, they maximize their profits:

$$\max \sum_{t=0}^{\infty} \beta^t \left(Z_t H_t N_t - P_{ft} Q_{ft} - P_{ot} Q_{ot} - W_t H_t N_t \right)$$

s.t.

$$H_{t+1} = (1-\delta) H_t + e_{ft}^{\alpha_f} Q_{ft}^{1-\alpha_f} + e_{ot}^{\alpha_o} Q_{ot}^{1-\alpha_o}$$

$$L_t + N_t + e_{ft} + e_{ot} = 1$$

Because firms confront the same constraints. Thus, we can obtain the same conclusion as before:

$$\frac{e_{ft}^{1-\alpha_f}}{e_{ot}^{1-\alpha_o}} = \frac{\alpha_f Q_{ft}^{1-\alpha_f}}{\alpha_o Q_{ot}^{1-\alpha_o}}$$

$$\frac{P_{ft}}{P_{ot}} = \frac{\frac{\left(\frac{1}{\alpha_f}-1\right) e_{ft}}{Q_{ft}}}{\frac{\left(\frac{1}{\alpha_o}-1\right) e_{ot}}{Q_{ot}}}$$

Therefore, no matter who bears the training expenses, our conclusion still exists. It illustrates the external validity of our conclusion.

Other Robustness Checks

Changing Clustering Level

To account for potential correlations in error terms at higher levels, we adjust the clustering level of standard errors. While the baseline regression used household-level clustering, we also explore clustering at the community and county levels to further validate the robustness of our results (Abadie et al. 2023). The results in Panel A of Table A2 show that clustering at higher levels still yields significantly positive coefficients, providing further evidence of the robustness of the findings.

Change Model Specification

Following Cohn et al. (2022), we explore the robustness of our results by using Poisson regression to address potential issues with the logarithmic transformation of the dependent variable. The results, displayed in column (1) in Panel B of Table A2 show that the coefficient for digital learning remains significantly positive at the 1% level, further confirming the robustness of our baseline findings.

Exclude the Impact of the Pandemic

To mitigate the potential confounding effects of the COVID-19 pandemic, we exclude data from 2020 and 2022 and re-estimate the model using only data from 2014 to 2018. The results, shown in column (2) in Panel B of Table A2, indicate that even after excluding pandemic-related data, digital learning continues to have a significant positive effect on income, with statistical significance at the 1% level. This suggests that the positive effects of digital learning are not driven by pandemic-related factors.

Add Interaction Fixed Effects

To address potential omitted variable bias, we introduce interaction fixed effects to control for unobserved variables. Columns (1), (2), and (3) in Panel C of Table A2 sequentially add interaction fixed effects for county-year, industry-year, and occupation-year to account for time-varying effects across these dimensions. Specifically, the county-year interaction fixed effects control for regional differences in digital infrastructure, while the industry-year and occupation-year effects account for variations in income based on occupation and industry. The results remain significantly positive, confirming the robustness and unbiased nature of our results.

TABLE A1 | The full results of baseline results.

Variables	(1) ln(Labor income)
Digital learning	0.069***
	(0.016)
Age	0.132***
	(0.017)
Age square	−0.002***
	(0.000)
Martial status	0.013
	(0.034)
Education level	0.014**
	(0.006)
Gender	0.875**
	(0.430)
Hukou	0.040
	(0.039)
Ethnicity	−0.089
	(0.140)
Family size	−0.034***
	(0.007)
Dependency ratio	−0.009
	(0.057)
Health status	−0.006
	(0.006)
Year fixed effect	Yes
Individual fixed effect	Yes
Constant	6.761***
	(0.711)
Observations	52,194
Adjusted R-squared	0.580

Note: The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

Post-Treatment Bias Checks

Including time-varying controls like health status and family characteristics in fixed-effects regressions may introduce some post-treatment bias if these are affected by digital learning. First, we examine whether the time-varying control variables are themselves affected by digital learning, such as health status and family characteristics. Specifically, we regress each time-varying control on digital learning. As shown in Columns (1), (2), and (3) in Panel D of Table A2, we do not find any statistically significant effects. Second, we re-estimated our main model after excluding these time-varying controls. As shown in Column (4) in Panel D of Table A2, the coefficient on digital learning remains positive and statistically significant, with virtually no change in magnitude. Taken together, these results indicate that including these time-varying controls does not materially affect our estimates.

Other Mechanism Analysis: Mitigating Information Asymmetry

The digital economy, through online platforms, significantly reduces the cost of information acquisition, enabling workers to access vast amounts of information quickly and efficiently. This process not only enhances the ability of workers to make informed decisions but also supplements traditional interpersonal networks, thereby mitigating information asymmetry in the labor market (Kuhn and Skuterud 2004). Such reductions in

TABLE A2 | Other robustness checks.

Panel A: Change the cluster level		
Variables	(1) Cluster at community level ln(Labor income)	(2) Cluster at county level ln(Labor income)
Digital learning frequency	0.063***	0.069***
	(0.020)	(0.019)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	6.961***	6.769***
	(0.901)	(0.922)
Observations	44,860	47,364
Adjusted R-squared	0.566	0.568

Panel B: Other ways		
Variables	(1) Poisson regression Labor income	(2) Exclude the epidemic ln(Labor income)
Digital learning frequency	0.063***	0.066***
	(0.017)	(0.024)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	6.425***	4.740***
	(0.570)	(0.807)
Observations	72,501	37,597
Adjusted R-squared/Pseudo R^2	0.6423	0.560

Panel C: Add fixed effects			
Variables	(1) ln(Labor income)	(2) ln(Labor income)	(3) ln(Labor income)
Digital learning	0.050***	0.028*	0.040*
	(0.017)	(0.017)	(0.024)
Controls	Yes	Yes	Yes
County#Year FE	Yes	Yes	Yes
Industry#Year FE	No	Yes	Yes
Occupation#Year FE	No	No	Yes
Year fixed effect	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes
Constant	7.049***	7.521***	8.869***
	(0.799)	(0.734)	(1.078)
Observations	46,000	45,991	30,261
Adjusted R-squared	0.576	0.600	0.602

TABLE A2 | (Continued)

Panel D: Post-treatment bias checks				
Variables	(1) Health status	(2) Family size	(3) Dependency ratio	(3) ln(Labor income)
Digital learning	0.017	0.006	−0.003	0.069***
	(0.020)	(0.015)	(0.002)	(0.016)
Controls	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes	Yes
Constant	3.813***	1.165*	−0.076	6.623***
	(0.796)	(0.680)	(0.066)	(0.712)
Observations	52,194	52,194	52,194	52,194
Adjusted R-squared	0.844	0.681	0.786	0.579

Note: The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

TABLE A3 | The results of CEM.

Variables	(1) ln(Labor income)
Digital learning	0.062**
	(0.026)
Controls	Yes
Year fixed effect	Yes
Individual fixed effect	Yes
Constant	7.888***
	(1.352)
Observations	18,940
Adjusted R-squared	0.554

Note: The standard errors are clustered at the household level and are reported in parentheses. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

information asymmetry can improve labor market outcomes by ensuring that workers secure wages that better reflect their abilities.

To empirically examine whether digital learning facilitates the use of the internet as a primary source of information, we analyze survey responses that specifically address this question.[6] The regression results, presented in column (1) in Table A4, show that the coefficient for digital learning is significantly positive at the 1% level. In economic terms, each standard deviation increase in digital learning is associated with a 27.5% increase in the probability of using the internet as a key information source.

Next, we examine whether this reduction in information asymmetry can compensate for the limitations of offline interpersonal networks. Drawing on the concept of "guanxi income,"[7] which reflects the strength of offline interpersonal relationships, we test whether digital learning serves as a substitute for strong interpersonal networks in reducing information asymmetry. When guanxi income is zero, it signifies weak relationships, while non-zero values indicate stronger relationships. The results, presented in columns (2) and (3) in Table A4, reveal that digital learning significantly mitigates information asymmetry for workers in

TABLE A4 | Alleviating information asymmetry.

Variables	(1) All	(2) Weak relationships	(3) Strong relationships
	Internet as an important source of information		
Digital learning	0.684***	0.739***	0.264
	(0.043)	(0.049)	(0.342)
Controls	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes
Constant	4.603***	5.006***	−14.494
	(1.228)	(1.218)	(14.436)
Observations	26,815	23,288	2946
Adjusted R-squared	0.557	0.548	0.587

Note: The regression samples are the full sample, subsets with weak and strong interpersonal relationships in columns (1), (2) and (3), respectively. They are divided according to the median of interpersonal relationships. All regressions control for control variables, year fixed effects, and individual fixed effects. The standard errors in the first three columns are clustered at the household level. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

the weak relationship group but has no significant effect on those with strong relationships. This finding supports our hypothesis that digital learning serves as an important tool for accessing critical information, particularly for workers with limited offline social networks, thereby alleviating information asymmetry.

Other Heterogeneity Analysis

In this section, we explore the heterogeneity of the impact of digital learning on labor market outcomes, particularly in contrast to traditional offline learning. Digital learning, which is facilitated through the internet, offers significant advantages over traditional learning, which often requires prolonged offline participation. In contrast to traditional education, digital learning is not subject to the same geographical, temporal, or resource constraints. To investigate these differences, we conduct subgroup analyses based on urban–rural disparities, terrain characteristics, and climate conditions.

First, the unequal distribution of educational resources between urban and rural areas is a well-documented issue (Li et al. 2024). Rural individuals, particularly children, face significant barriers in accessing quality educational resources. Similarly, rural adults encounter difficulties in obtaining offline vocational training, limiting opportunities for human capital development. To test the impact of digital learning in this context, we perform group regressions based on urban–rural heterogeneity. The results, presented in Panel A of Table A5, indicate that digital learning has a significant positive effect on the income of rural laborers, while there is no significant effect for urban laborers. This finding aligns with our earlier analysis, suggesting that digital learning, by overcoming geographical and resource constraints, alleviates information asymmetry, facilitates human capital accumulation, and promotes non-agricultural employment and income growth in rural areas.

Second, geographical conditions play a crucial role in limiting access to traditional learning, particularly in remote regions where transportation difficulties pose significant barriers. To examine whether digital learning can overcome these geographical constraints, we divide the sample based on the median terrain ruggedness. The results in Panel B of Table A5 show that digital learning significantly increases the income of individuals residing in more rugged terrain, but has no significant effect on those in less rugged terrain. This suggests that digital learning can mitigate the challenges posed by difficult terrain, help alleviate information asymmetry, foster human capital development, and ultimately improve labor

TABLE A5 | Other heterogeneous effect of digital learning.

Panel A: Heterogeneity by urban–rural

Variables	(1) Urban area ln(Labor income)	(2) Rural area ln(Labor income)
Digital learning	0.007	0.085***
	(0.026)	(0.021)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	6.630**	6.760***
	(2.843)	(0.807)
Observations	10,744	41,450
Adjusted R-squared	0.577	0.553

Panel B: Heterogeneity by terrain region

Variables	(1) Gentle terrain region ln(Labor income)	(2) Steep terrain region ln(Labor income)
Digital learning	0.031	0.098***
	(0.023)	(0.023)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	5.145***	7.689***
	(1.110)	(0.938)
Observations	25,118	26,400
Adjusted R-squared	0.572	0.578

Panel C: Heterogeneity by climate

Variables	(1) Low climate risk area ln(Labor income)	(2) High climate risk area ln(Labor income)
Digital learning	0.034	0.103***
	(0.033)	(0.028)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	6.124***	7.366***
	(1.138)	(1.357)
Observations	26,061	25,558
Adjusted R-squared	0.558	0.577

Note: In Panel A, we report the heterogeneity by urban–rural. The regression samples live in the urban area in column (1) and live in the rural area in column (2). In Panel B, we report the heterogeneity by terrain region. The regression samples live in the gentle terrain region in column (1) and live in the steep terrain region in column (2). It is divided according to the median of terrain relief. In Panel C, we report the heterogeneity by climate. The regression samples live in the low climate risk area in column (1) and live in the high climate risk area in column (2). It is divided according to the median of climate risk rate. All regressions control for control variables, year fixed effects, and individual fixed effects. The standard errors in the first three columns are clustered at the household level. The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

TABLE A6 | Heterogenous effect by income.

Panel A: Heterogeneity by income		
Variables	**(1)** **Low income** **ln(Labor income)**	**(2)** **High income** **ln(Labor income)**
Digital learning	0.115***	0.024***
	(0.036)	(0.008)
Controls	Yes	Yes
Year fixed effect	Yes	Yes
Individual fixed effect	Yes	Yes
Constant	7.414***	9.676***
	(0.998)	(0.509)
Observations	27,933	24,261
Adjusted R-squared	0.718	0.796

Panel B: Quantile regression			
Variables	**(1)** **25%** **ln(Labor income)**	**(2)** **50%** **ln(Labor income)**	**(3)** **75%** **ln(Labor income)**
Digital learning	0.161***	0.080	0.061
	(0.042)	(0.098)	(0.605)
Controls	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Individual fixed effect	Yes	Yes	Yes
Constant	6.630**	6.760***	6.760***
	(2.843)	(0.807)	(0.807)
Observations	52,194	52,194	52,194
Number of groups	21,162	21,162	21,162

Note: The significance levels of 1%, 5%, and 10% are denoted by ***, **, and *, respectively.

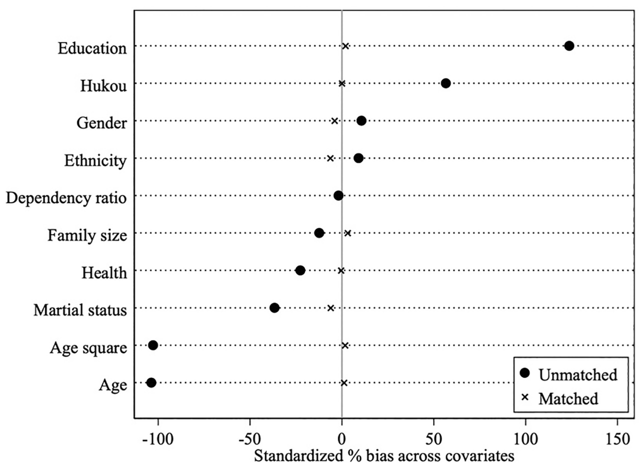

FIGURE A1 | Balance test for PSM. The solid dots and cross signs represent the standardized bias of covariates between digitally learned and non-digitally learned samples before and after matching, respectively.

market performance, particularly in areas where traditional learning is harder to access.

Finally, climate conditions also affect the feasibility and effectiveness of traditional learning. For instance, extreme weather events often result in the cancellation of in-person learning sessions. To test the impact of climate on digital learning, we use provincial-level climate risk data from Guo et al. (2024) and divide the sample based on the median level of climate risk. The results in Panel C of Table A5 show that digital learning has a significant positive effect on the income of individuals in high climate-risk provinces, while there is no significant effect for those in low-risk regions. This suggests that digital learning is resilient to extreme weather conditions, which can disrupt traditional learning opportunities.

In summary, the heterogeneity analysis demonstrates that digital learning offers distinct advantages over traditional offline learning by overcoming the uneven distribution of educational resources, geographical constraints, and the impacts of adverse weather conditions. Thus, digital learning complements traditional learning by addressing these barriers and jointly promoting the accumulation of human capital. It helps alleviate labor market imbalances, particularly in underserved regions, by enhancing income and employment opportunities for individuals facing resource limitations.